Absolute value equations

1

a $|x| = 9$

b $|5x - 2| = 3$

2

a 2 solutions

b There are no solutions.

3 $b = 0$

4 Neither, because the equation has no solutions since the the left hand side of the equation can never be negative.

5

a $x = \pm 8$

b $x = \pm 10$

c $x = \pm \frac{4}{9}$

d $x = \pm 5$

e $x = \pm 40$

f $x = \pm 42$

g $x = \pm 6$

h $x = \pm \frac{2}{5}$

6

a $x = 4, -10$

b $x = 1, -11$

c $x = -8, 14$

d $x = -2, 4$

e $x = -8, 22$

f $x = -6, 14$

7

a $x = 2, -8$

b $x = 1, -5$

c $x = -2, -8$

d $x = 1, -9$

e $x = -\frac{1}{3}, 5$

f $x = -\frac{8}{5}, 2$

g $x = -4, \frac{2}{3}$

h $x = -1, -\frac{11}{4}$

8

a $x = \pm \frac{9}{2}$

b $x = \pm 7$

c $x = \pm \frac{1}{2}$

d $x = 5, -1$

e $x = \pm 5$

f $x = -3, -5$

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